

Linux & DevOps Practical Implementation on Ubuntu

Objective:

The objective of this task is to perform and understand essential Linux and DevOps operations on an Ubuntu system. This includes user management, file handling, permissions, process management, package installation, server setup (Apache, MySQL, Tomcat), and text processing using tools like AWK and SED.

Prerequisites:

- Ubuntu 24.04 / 25.04 installed (VM or physical system)
- User with sudo/root privileges
- Internet connection (for package installation)
- Basic knowledge of Linux terminal commands

Implementation Steps:

1) Create a new user:

Ans: Open terminal and Create user and Enter password then successfully created to check id username then enter we get details of new user like id, groups, owners.

```
vboxuser@Arunsa:~$ id techie
uid=1003(techie) gid=1003(techie) groups=1003(techie)
vboxuser@Arunsa:~$ id Arun
uid=1001(Arun) gid=1001(Arun) groups=1001(Arun)
```

2) Navigate to the home directory.

Ans: Open terminal and use the command to change directory to home (cd ~).

```
vboxuser@Arunsa:~$ ~
bash: /home/vboxuser: Is a directory
vboxuser@Arunsa:~$ cd ~
vboxuser@Arunsa:~$
```

3) Create a new directory.

Ans: mkdir directory name to create new directory.

```
vboxuser@Arunsa:~$ mkdir Devops
vboxuser@Arunsa:~$ ls
alphabet  Desktop  Documents  Music      Public      snap      test      Videos
arun.tar  Devops   Downloads  Pictures   sample.war  Templates  test_tar
```

4) List the contents of a directory.

Ans: Use the command (ls) to display the contents of the current directory.

```
vboxuser@Arunsa:~$ ls
alphabet  Desktop  Documents  Music      Public      snap      test      Videos
arun.tar  Devops   Downloads  Pictures   sample.war  Templates  test_tar
```

5) Change the current directory.

Ans: Use the command `cd directory_name` to change the current directory.

```
vboxuser@Arunsaï:~$ cd Devops
vboxuser@Arunsaï:~/Devops$
```

6) Create a new empty file.

Ans: Use the command `touch file_name` to create a new empty file.

```
vboxuser@Arunsaï:~/Devops$ touch desk
vboxuser@Arunsaï:~/Devops$ ls
desk
```

7) View the contents of a file.

Ans: Use the command `cat file_name` to display the contents of a file.

```
vboxuser@Arunsaï:~/Devops$ cat desk
hello world
techie
vboxuser@Arunsaï:~$ cat test
```

8) Copy a file to another location.

Ans: Use the command `cp source_file destination_path` to copy a file.

```
vboxuser@Arunsaï:~/Devops$ cp desk /tmp/
vboxuser@Arunsaï:~/Devops$
```

9) Move a file to another location.

Ans: The file "desk" is successfully found in the /tmp directory using the `ls` command.

```
vboxuser@Arunsaï:~/Devops$ ls
vboxuser@Arunsaï:~/Devops$
```

```
vboxuser@Arunsaï:~/Devops$ ls /tmp
desk
```

10) Rename a file.

Ans: The file name changes from desk to new desk in the directory.

```
vboxuser@Arunsaï:~/Devops$ cd /tmp
vboxuser@Arunsaï:/tmp$ mv desk newdesk
vboxuser@Arunsaï:/tmp$ ls
newdesk
```

11) Delete a file.

Ans: Use the command `rm file_name` to delete a file.

```
vboxuser@Arunsai:/tmp$ rm newdesk
vboxuser@Arunsai:/tmp$ ls
snap-private-tmp
```

12) Grant or revoke permissions on a file or directory.

Ans: Use the command `chmod 755 Devops` to change permissions.

```
vboxuser@Arunsa:~$ chmod 755 Devops
vboxuser@Arunsa:~$ ls -l
total 72
-rw-rw-r-- 1 vboxuser vboxuser    36 Mar 27 10:00 alphabet
-rw-rw-r-- 1 vboxuser vboxuser 10240 Mar 28 07:52 arun.tar
drwxr-xr-x 6 vboxuser vboxuser  4096 Mar 27 06:13 Desktop
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 29 01:33 Devops
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Documents
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Downloads
```

13) View the current date and time.

Ans: Use the command date to display the current date and time.

```
vboxuser@Arunsa:~$ date
Sun Mar 29 02:05:09 AM UTC 2026
vboxuser@Arunsa:~$
```

14) Check the system uptime.

Ans: Use the command uptime to check how long the system has been running.

```
vboxuser@Arunsai:~$ uptime
02:06:28 up 16:27, 1 user, load average: 0.00, 0.03, 0.00
vboxuser@Arunsai:~$
```

15) View the running processes.

Ans: Use the command `ps aux` to display all running processes in the system.

```
vboxuser@Arunsai:~$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.4	23180	8328	?	Ss	Mar28	0:21	/sbin/init sp

16) Kill a running process.

Ans: Use the command kill process_id to terminate a running process or Kill -9 process id.

```
vboxuser@Arunsa:~/Desktop$ ps -ef | grep nginx
root      9582      1  0 07:34 ?        00:00:00 nginx: master process /usr/sbin/nginx -g daemon o
master_process on;
www-data  9612    9582  0 07:37 ?        00:00:00 nginx: worker process
vboxuser  9621    5077  0 07:37 pts/1    00:00:00 grep --color=auto nginx
vboxuser@Arunsa:~/Desktop$ sudo kill -9 9582
```

17) Install a package using the package manager (e.g., apt or yum).

Ans: Use the command sudo apt install nginx to install a package

```
vboxuser@Arunsa:~/Desktop$ sudo apt install nginx

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.6).
0 upgraded, 0 newly installed, 0 to remove and 41 not upgraded.
vboxuser@Arunsa:~/Desktop$
```

18) Update the system packages.

Ans: Use sudo apt update && sudo apt upgrade -y to update packages.

```
vboxuser@Arunsa:~$ sudo apt update && sudo upgrade -y

Get:1 file:/cdrom/noble InRelease
Ign:1 file:/cdrom/noble InRelease
Get:2 file:/cdrom/noble Release
Err:2 file:/cdrom/noble Release
  File not found - /cdrom/dists/noble/Release (2: No such file or directory)
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
```

19) Create a symbolic link.

Ans: Use the command ln -s target_file link_name to create a symbolic link.

```
vboxuser@Arunsa:~$ ln -s desk link_desk
vboxuser@Arunsa:~$ ls -l
total 72
-rw-rw-r-- 1 vboxuser vboxuser   36 Mar 27 10:00 alphabet
-rw-rw-r-- 1 vboxuser vboxuser 10240 Mar 28 07:52 arun.tar
drwxr-xr-x 6 vboxuser vboxuser  4096 Mar 27 06:13 Desktop
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 29 01:33 Devops
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Documents
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Downloads
lrwxrwxrwx 1 vboxuser vboxuser    4 Mar 29 02:55 link_desk -> desk
```

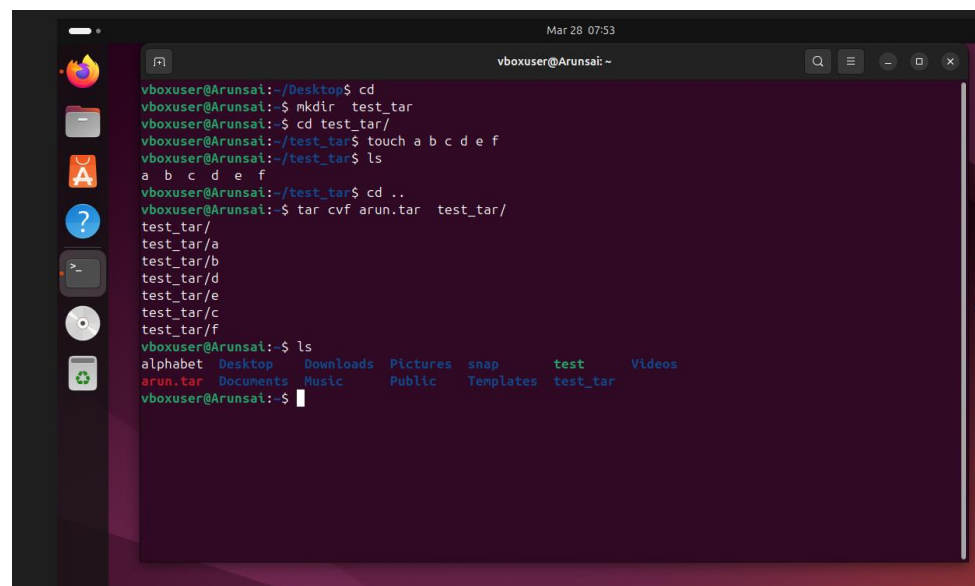
20) Search for files using the find command.

Ans: Use the command `find path -name file_name` to search for files in a directory.

```
vboxuser@Arunsaï:/$ find / -name *.txt
/etc/brltty/Input/vr/all.txt
/etc/brltty/Input/vs/all.txt
/etc/brltty/Input/tt/all.txt
/etc/brltty/Input/bl/40_m20_m40.txt
/etc/brltty/Input/bl/18.txt
```

21) Compress and decompress files using tar.

Ans: Use the command `tar -cvf archive_name.tar folder_name` to compress the folder.



```
Mar 28 07:53
vboxuser@Arunsaï:~
vboxuser@Arunsaï:~/Desktop$ cd
vboxuser@Arunsaï:~$ mkdir test_tar
vboxuser@Arunsaï:~$ cd test_tar/
vboxuser@Arunsaï:~/test_tar$ touch a b c d e f
vboxuser@Arunsaï:~/test_tar$ ls
a b c d e f
vboxuser@Arunsaï:~/test_tar$ cd ..
vboxuser@Arunsaï:~$ tar cvf arun.tar test_tar/
test_tar/
test_tar/a
test_tar/b
test_tar/d
test_tar/e
test_tar/c
test_tar/f
vboxuser@Arunsaï:~$ ls
alphabet Desktop Downloads Pictures snap test Videos
arun.tar Documents Music Public Templates test_tar
vboxuser@Arunsaï:~$
```

22) Monitor system resources with top or htop.

Ans: Use the command `top` to monitor system resources in real-time.

```
vboxuser@Arunsaï:/$ top
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2521	vboxuser	20	0	3593220	325424	38344	S	1.0	16.2	15:15.45	gnome-shell

23) Create and manage user groups.

Ans: Create a group and add a user to the group.

- `groupadd devteam`
- `usermod -aG devteam techie`

```
vboxuser@Arunsaï:/$ groups techie
techie : techie devteam
vboxuser@Arunsaï:/$
```

24) Set up SSH password less authentication.

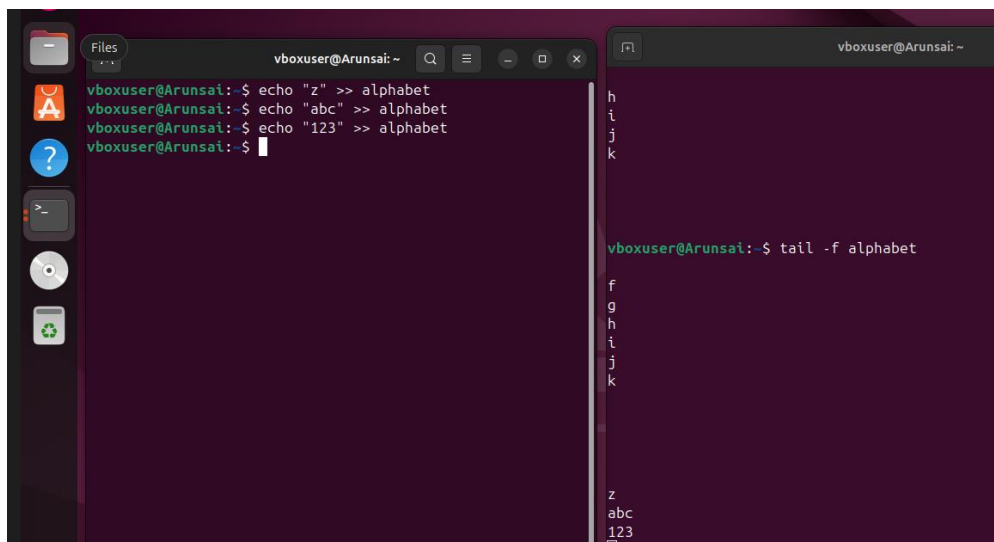
Ans: SSH login works without password.

- Generate SSH key using ssh-keygen.
- Accept default file path.
- Copy public key to remote system using ssh-copy-id.

```
boxuser@Arunasai:/$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/vboxuser/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vboxuser/.ssh/id_ed25519
Your public key has been saved in /home/vboxuser/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:DuHZ/DZf5Q5AgcIf+by+Ns4l8+ww8x88IooRPVeFiKg vboxuser@Arunasai
The key's randomart image is:
--[ED25519 256]--+
  .. .oo .. |
  .o.+ . o. |
  o o +.. |
  E =. ..+ |
  +.So ... .|
  o..o ...o |
  .. +oo.++.|
  o..+=B+oo|
  . . o=++=*|
  -----[SHA256]-----+
```

25) Monitor log files using tail or grep.

Ans: Use tail -f file_name to monitor file updates in real-time and Modify the file to observe live changes.

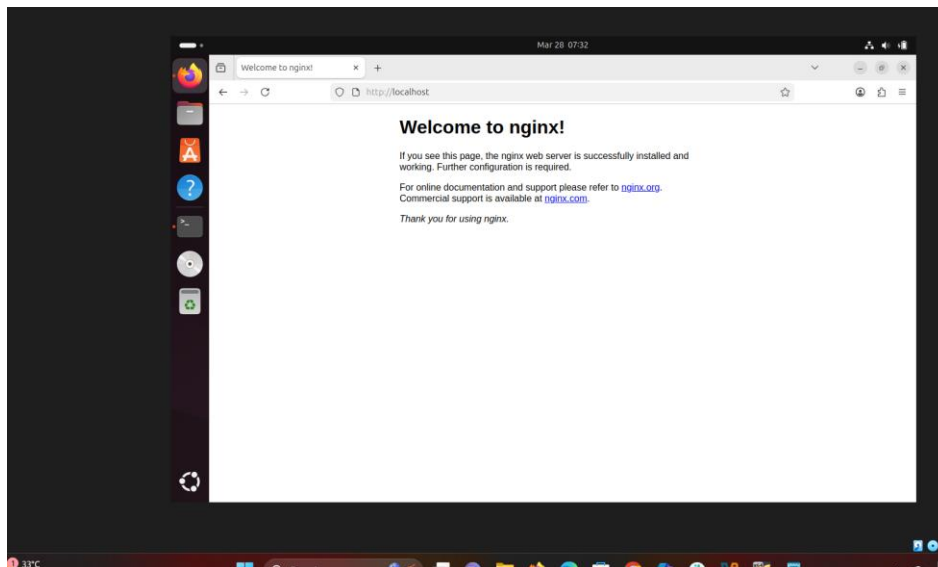


26) Set up a web server (e.g., Apache or Nginx).

1. Install Nginx using package manager.
2. Start and enable the Nginx service.
3. Access the web server using browser

```
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.6).
0 upgraded, 0 newly installed, 0 to remove and 41 not upgraded.
vboxuser@Arunasai:~/Desktop$ sudo apt install nginx
(etc/passwd:12:?) syntax error

root      6622      2  0 06:26 ?        00:00:00 [kworker/0:1H-kblockd]
root      7498      2  0 06:40 ?        00:00:01 [kworker/u4:5-flush-8:0]
root      8002      2  0 06:55 ?        00:00:00 [psimon]
root      8079      2  0 06:55 ?        00:00:00 [kworker/0:0H-kblockd]
root      8186      1  0 07:08 ?        00:00:00 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
www-data  8187      8186  0 07:08 ?        00:00:00 nginx: worker process
root      8195      2  0 07:10 ?        00:00:00 [kworker/0:2-cgroup_free]
root      8225      2  0 07:15 ?        00:00:00 [kworker/u4:1-events_unbound]
root      8232      2  0 07:19 ?        00:00:00 [kworker/u5:2]
root      8243      2  0 07:20 ?        00:00:00 [kworker/0:2H-kblockd]
```



27) Configure and secure a MySQL Database.

Ans: Steps to Secure MySQL Database

- Install MySQL server.
- Start and enable MySQL service.
- Configure security settings.

```
vboxuser@Arunasai:/$ sudo apt install mysql-server
```

```
vboxuser@Arunasai:~$ sudo systemctl start mysql
vboxuser@Arunasai:~$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
```

```
vboxuser@Arunasai:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-03-29 05:17:07 UTC; 3min 20s ago
     Main PID: 4438 (mysqld)
    Status: "Server is operational"
      Tasks: 38 (limit: 2261)
     Memory: 353.4M (peak: 367.9M)
        CPU: 4.612s
    CGroup: /system.slice/mysql.service
            └─4438 /usr/sbin/mysqld

Mar 29 05:17:05 Arunsai systemd[1]: Starting mysql.service - MySQL Community Server.
Mar 29 05:17:07 Arunsai systemd[1]: Started mysql.service - MySQL Community Server.
Lines 1-13/13 (END)
```

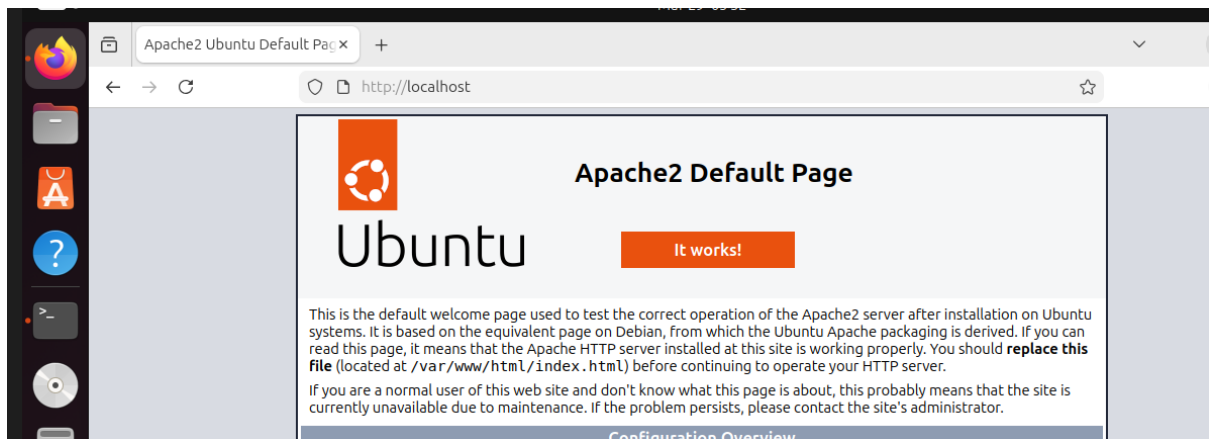

28) Set up a Application Server (e.g.,Apache Tomcat)

Ans: Steps:

- Install Apache server.
- Start and enable the service.
- Access the server through browser.

```
vboxuser@Arunsa:~$ sudo systemctl start apache2
vboxuser@Arunsa:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
   Active: active (running) since Sun 2026-03-29 05:29:05 UTC; 4s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 6118 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU>
   Main PID: 6134 (apache2)
    Tasks: 55 (limit: 2261)
   Memory: 5.4M (peak: 5.5M)
      CPU: 103ms
   CGroup: /system.slice/apache2.service
           └─6134 /usr/sbin/apache2 -k start
             └─6136 /usr/sbin/apache2 -k start
               └─6137 /usr/sbin/apache2 -k start

Mar 29 05:29:05 Arunsa systemd[1]: Starting apache2.service - The Apache HTTP >
Mar 29 05:29:05 Arunsa apachectl[6133]: AH00558: apache2: Could not reliably d>
Mar 29 05:29:05 Arunsa systemd[1]: Started apache2.service - The Apache HTTP S>
lines 1-17/17 (END)
```



29) create a service file for Apache Tomcat.(Should execute by using systemctl command)

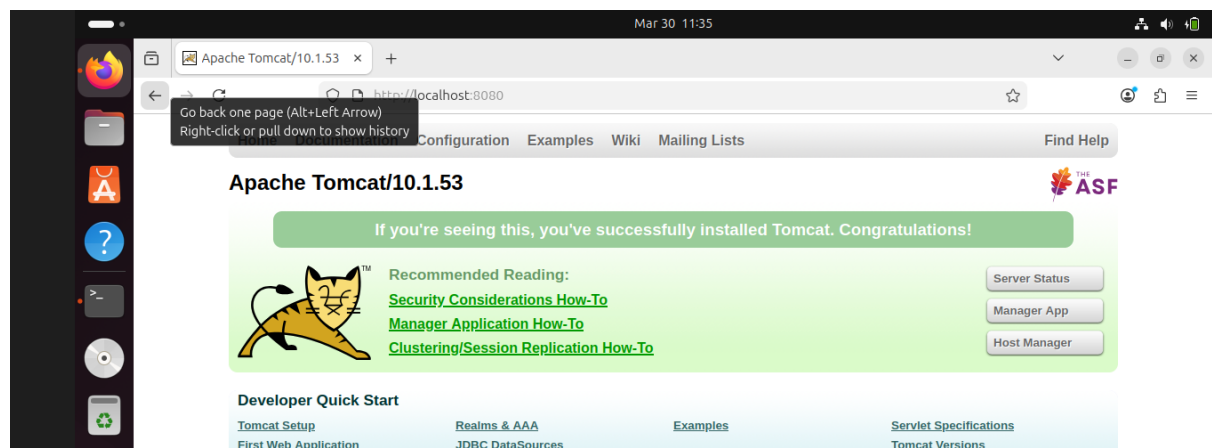
Ans: Steps:

- 1.Install Java
- 2.Downloaded Apache Tomcat
- 3.Extract Tomcat
- 4.Set Permissions

5. Create Systemd Service File
6. Reload Systemd and start service
7. Enable Auto Start

```
vboxuser@Arunsa: /opt/tomcat/bin$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: enabled)
   Active: active (running) since Mon 2026-03-30 12:08:14 UTC; 1s ago
     Process: 7959 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
    Main PID: 7966 (java)
      Tasks: 13 (limit: 2261)
     Memory: 41.0M (peak: 41.0M)
        CPU: 871ms
    CGroup: /system.slice/tomcat.service
            └─7966 /usr/lib/jvm/java-11-openjdk-amd64/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties

Mar 30 12:08:14 Arunsai systemd[1]: tomcat.service: Deactivated successfully.
Mar 30 12:08:14 Arunsai systemd[1]: tomcat.service: Consumed 7.240s CPU time.
Mar 30 12:08:14 Arunsai systemd[1]: tomcat.service: Scheduled restart job, restart counter is at 0.
Mar 30 12:08:14 Arunsai systemd[1]: Starting tomcat.service - Apache Tomcat...
Mar 30 12:08:14 Arunsai startup.sh[7959]: Tomcat started.
Mar 30 12:08:14 Arunsai systemd[1]: Started tomcat.service - Apache Tomcat.
lines 1-17/17 (END)
vboxuser@Arunsa: /opt/tomcat/bin$ sudo systemctl enable tomcat
Created symlink /etc/systemd/system/multi-user.target.wants/tomcat.service → /etc/systemd/system/tomcat.service.
```



30) Print specific columns from a delimited file.

Ans: The awk command successfully prints selected columns from the file.
Only specified columns are displayed in the output.

```
vboxuser@Arunsaï:~$ cat test
id name city
1 Arun HYd
2 Mahi Che
3 Raj Ban
vboxuser@Arunsaï:~$ awk '{print $1, $2}' test
id name
1 Arun
2 Mahi
3 Raj
vboxuser@Arunsaï:~$ awk '{print $1, $2, $3}' test
id name city
1 Arun HYd
2 Mahi Che
3 Raj Ban
```

31) Filter and print lines based on a specific pattern or condition.

Ans: 1. Use grep to filter lines based on text pattern.
2. Use awk to filter lines based on conditions.

```
vboxuser@Arunsaï:~$ grep "raju" test
raju 60
vboxuser@Arunsaï:~$ awk ' $2 > 50' test
mahï 80
raju 60
vboxuser@Arunsaï:~$
```

32) Calculate and print the average, sum, or other statistics of a column.

Ans: Use awk to calculate sum and average of a numeric column.

- awk '{sum += \$2} END {print "Sum:", sum}' test
- awk '{sum += \$2; count += 1} END {print "Average:", sum / count}' test

```
vboxuser@Arunsaï:~$ awk '{sum += $2} END {print "Sum:", sum}' test
Sum: 180
vboxuser@Arunsaï:~$
```

```
Average: 60
vboxuser@Arunsaï:~$ awk '{sum += $2; count += 1} END {if (count > 0) print "Average:", sum / count}' test
Average: 60
vboxuser@Arunsaï:~$
```

33) Perform string manipulation, such as extracting substrings or changing case.

Ans: The sed command successfully replaces specific strings in the file.

```
vboxuser@Arun Sai:~$ cat test
mahi 80
raju 60
sai 40
vboxuser@Arun Sai:~$ sed 's/mahi/MAHI/' test
MAHI 80
raju 60
sai 40
vboxuser@Arun Sai:~$ sed 's/raju/RAJU/' test
mahi 80
RAJU 60
sai 40
vboxuser@Arun Sai:~$
```

34) Count the occurrences of a specific pattern in a file.

Ans: Use `grep -c "pattern" file` to count occurrences.

```
vboxuser@Arun Sai:~$ cat > test
mahi
raju
sai prasad
raju
vboxuser@Arun Sai:~$ cat test
mahi
raju
sai prasad
raju
vboxuser@Arun Sai:~$ grep c "raju" test
grep: raju: No such file or directory
vboxuser@Arun Sai:~$ grep -c "raju" test
2
vboxuser@Arun Sai:~$ grep -c "mahi" test
1
```

35) Sort lines based on a specific field or column.

Ans: Use the `sort` command with column options to sort data based on a specific field.

```
vboxuser@Arun Sai:~$ cat > test
kumar
sai
ganesh
mahi
vboxuser@Arun Sai:~$ sort -k1 test
ganesh
kumar
mahi
sai
```

36) Merge multiple files based on a common field or column.

Ans: 1. Create two files with a common column.
 2. Sort both files.
 3. Use `join` command to merge them.

```
vboxuser@Arunsa:~$ cat > file1
1 mahi
2 raju
3 sai
vboxuser@Arunsa:~$ cat > file2
1 80
2 40
3 20
vboxuser@Arunsa:~$ join file1 file2
1 mahi 80
2 raju 40
3 sai 20
```

37) Substitute text in a file using search and replace.

Ans: Use the sed command to search and replace text in a file.

```
vboxuser@Arunsa:~$ sed 's/a/A/' test
kumAr
sAi
gAnesh
mAhi
vboxuser@Arunsa:~$
```

38) Delete specific lines based on a pattern or line number.

Ans: Use the sed command to delete lines either by pattern or by line number.

```
vboxuser@Arunsa:~$ cat test
kumar
sai
ganesh
mahI
vboxuser@Arunsa:~$ sed 2d test
kumar
ganesh
mahI
vboxuser@Arunsa:~$
```

39) Append or insert text before or after a specific pattern or line.

Ans: Steps for before and after text pattern

- Use sed '2i text' file to insert before a line.
- Use sed '/pattern/a text' file to append after a pattern.

```
vboxuser@Arunsa:~$ cat test
kumar
sai
ganesh
mahi
vboxuser@Arunsa:~$ sed '2i rajesh' test
kumar
rajesh
sai
ganesh
mahi
vboxuser@Arunsa:~$ sed '/ganesh/a gopi' test
kumar
sai
ganesh
gopi
mahi
```

40) Print only specific lines from a file.

Ans: In two ways for specific line to print

Use sed -n 'line_numberp' file to print a specific line.

Use sed -n 'start, endp' file to print a range of lines.

```
vboxuser@Arunsa:~$ cat test
kumar
sai
ganesh
mahi
vboxuser@Arunsa:~$ sed -n '2p' test
sai
```

41) Copy file from linux to windows machine

Ans: From Windows to Linux and same way Linux to windows through gitbash


```
vboxuser@Arunasai:~$ awk '{sum+=$2} END {print sum}' test
0
```

4. Print specific columns

- Extract particular columns from a file
awk '{print \$1, \$2}' file

```
vboxuser@Arunasai:~$ awk '{print $1, $2}' test
kumar
sai
ganesh
mah
```

5. Perform calculations

- Calculate sum, average, etc.
awk '{sum+=\$2} END {print sum}' file

```
vboxuser@Arunasai:~$ sed '2a babu' file1
1 mahi
2 raju
babu
3 sai
```